DAISY FIELD | OXI ONE MK1 | SYNTHEX Project

Optimal Drum Machine

UX / UI Design

Hybrid architecture — Daisy Field as sound & performance surface
OXI One MK1 as sequencing brain · complete 3-layer control map

16
addressable controls

3×
parameter layers

DaisySP · libDaisy · MIDI Clock Sync · 3-State LED Encoding

AGENDA

4 SECTIONS

01

Core UX Concepts

Drum machine design pillars, control paradigm taxonomy, immediacy principles

02

OXI One as Sequencing Brain

Function offloading, freed Daisy Field resources, MIDI clock integration

03

Daisy Field Implementation

3-state LED encoding, control layers, OLED architecture, firmware patterns

04

Full Control Map

Master parameter table, information density analysis, design rationale

SECTION 01

Core UX Concepts of Drum Machines

01

What separates immediate, performable hardware from menu-buried disappointments — the design principles every drum machine UI is evaluated against.

THE 5 PILLARS OF DRUM MACHINE UX

1

IMMEDIACY

Zero-menu access to the most-used parameters. Volume, pitch, decay reachable in one gesture without mode changes. Cognitive latency from intention to action must be negligible.

2

VISUAL STATE

Every active state visible at a glance. Step-on LEDs, mute state, playback position, clock phase — all encoded in hardware. No screen navigation to read machine state.

3

LAYER ECONOMY

Second functions via hold/shift, not nested menus. Each additional layer doubles parameter space at zero hardware cost. Maximum 2 layers before muscle-memory breaks down.

4

TRANSPORT LOCK

Play/stop/record reachable without mode changes. BPM, pattern advance, fill always available during performance. No state machine dead-ends that require stopping to navigate.

5

VOICE FOCUS

One control context at a time. Selecting a voice reassigns all knobs to that voice immediately. Eliminates confusion of shared encoders with unclear selection state.

CONTROL PARADIGM TAXONOMY

Three paradigms — cognitive load & performance profile

Paradigm	Device	Step Entry	Param Access	Visual Feedback	Daisy Field fit
Step-button	Roland TR-8S	Dedicated 16-step row	Knob-per-voice column	Step LEDs always-on	Low — paradigm mismatch: no step row
Pad / Groovebox	Novation Circuit	Pad grid dual-use	8 macro knobs + shift	RGB pad color coding	Med — closest form factor match
Encoder matrix	Elektron M:Samples	Trig key row	6 encoders + FUNC	OLED + trig LEDs	Low — OLED UX pattern useful
Hybrid (this design)	Daisy + OXI One	Offloaded to OXI	8 knobs × 3 layers	3-state key LEDs	Native — built for this target

SECTION 02

OXI One as the Sequencing Brain

Which functions the OXI One absorbs — and precisely what that liberates on the Daisy Field control surface.

OXI ONE — FUNCTION OFFLOADING MAP

OXI ONE HANDLES	DAISY FIELD — FREED FOR:
Step Sequencing DRUM mode — 8 gate outs, 16 steps per track, pattern chain, per-step probability	All 16 keys become performance + status display (no step entry role)
Transport Play / Stop / Record, start/stop triggers, tap tempo encoder	SW1 / SW2 freed from transport → available for 2× layer expansion
Pattern Management 20 projects × 16 patterns, arrangement/chain mode	Row A = voice mute/solo/select with persistent 3-state LED encoding
Clock Distribution MIDI clock 24ppq → Daisy MIDI IN, sync pulse output	Row B = contextual: FX routing, pattern queue, step on/off editing
MIDI Note Triggers Note On per voice hit per step → Field receives for LED flash	Trigger flash per Note On = real-time voice activity indicator
CV / Gate Modulation Mod lanes → OXI CV OUT → Daisy Field CV IN 1–4	CV IN 1–4 = modulation targets driven by OXI modulation lanes

SECTION 03

Daisy Field Implementation

Complete UX architecture for the Daisy Field surface:
3-state LED system · control layers · OLED strategy · MIDI clock integration

HARDWARE INVENTORY
8× knobs K1–K8 + 9th hardware volume
16× keys with LED (Row A + Row B)
2× switches SW1, SW2
128×64 OLED display
CV IN 1–4 · CV OUT 1–2 · Gate IN/OUT
MIDI IN / MIDI OUT (TRS 3.5mm)

3-STATE LED ENCODING SYSTEM · 128×64px OLED FIELD

OFF

Slot empty / not in kit

Voice not assigned in current kit. Not sent by OXI in DRUM mode. No CPU overhead.

- Empty FX slot
- Unused voice position
- Pattern slot void

BLINK

Active but muted

OXI pattern running, voice silenced on Daisy. Blink phase toggles at 16th-note rate via MIDI clock 24ppq.

- Muted during performance
- Queued pattern (next)
- FX bypass active

ON

Active and playing

Voice unmuted, playing steps. Solid-on between trigger flashes. Selected track in Row A.

- Active voice playing
- Selected track (Row A)
- Current active pattern

FLASH *

Trigger hit — 80ms

Momentary override on Note On from OXI. Brightens to full for 80ms, returns to persistent LED state.

- Beat accent indicator
- Performance hit confirm
- OXI step trigger

* FLASH is software-generated: duration distinguishes it from sustained ON. Driven exclusively by OXI MIDI Note On messages. No additional GPIO required.

CONTROL MAPPING — NORMAL PLAY LAYER

SW1 released · SW2 released

K1–K8 · VOICE PARAMETERS

K1	**LEVEL**	Voice amplitude 0–100%
K2	**PITCH**	Tune / fundamental frequency
K3	**DECAY**	Envelope decay time
K4	**PAN**	Stereo position L–R
K5	**FILTER**	Low-pass filter cutoff
K6	**REVERB**	Reverb send level
K7	**DELAY**	Delay send level
K8	**PROB**	Step probability 0–100%

KEY ROWS · TRACK + STEP CONTROL

ROW A — VOICE SELECT (tap to select, knobs follow)

BD	SD	CH	OH	CP	TM	RM	FX

LED: ON = active/unmuted · BLINK = muted · OFF = empty · Bright-yellow = selected track

ROW B — STEP ON/OFF (steps 1–8 · SW2 tap = page 2 → steps 9–16)

1	2	3	4	5	6	7	8

Active step = yellow · Inactive step = grey · Tap to toggle. OXI reads the resulting pattern via Gate OUT.

SW1 hold → shift layer (knobs + Row A reassign) · **SW2** tap → page flip 1–8 ↔ 9–16 · **SW2** hold → pattern select
SW1 + SW2 simultaneously → save current pattern

CONTROL MAPPING — SW1 SHIFT LAYER

Hold SW1

Holding SW1 instantly reassigns K1–K8 and Row A from voice-centric to global/transport. Row B retains step 9–16 page role.

K1–K8 · GLOBAL / TRANSPORT

K1	BPM	Master tempo — live override of OXI clock master
K2	SWING	Global shuffle amount 0–100%
K3	MASTER VOL	Master output gain pre-DAC
K4	PAT LEN	Active pattern step length 1–16
K5	FILL PROB	Auto-fill trigger probability
K6	CLK DIV	Internal clock divider for slow voices
K7	GATE LEN	Global gate length override

ROW A · MUTE / UNMUTE

BD	SD	CH	OH	CP	TM	RM	FX

Tap while holding SW1 → toggle mute.
BLINK = muted (OXI running, Daisy silenced).
ON = active and passing audio.

ROW B · STEPS 9–16

9	10	11	12	13	14	15	16

Step row shifts to 9–16 context.
Toggle on/off same mechanics as normal layer.

Design principle: shift accessed by physical hold, not mode toggle — eliminates mode-lock. Release SW1 = instant return. OLED lines 3–4 update in real time to show shift parameter names.

SW2 — STEP PAGE TOGGLE & PATTERN SELECT MODE

Tap · Hold

SW2 TAP · STEP PAGE FLIP (1–8 ↔ 9–16)

Each voice holds 16 steps. Row B (8 keys) displays one page of 8. SW2 tap flips pages. OLED Line 2 shows active page indicator.

PAGE 1 (steps 1–8) → SW2 tap → PAGE 2 (steps 9–16)

1	2	3	4	5	6	7	8

Active page

9	10	11	12	13	14	15	16

After SW2 tap

OXI One sequences both pages independently. Daisy displays only the page being edited. The OXI step cursor position is shown on OLED Line 2 — freeing Row B purely for toggling, not for displaying playback position.

SW2 HOLD · PATTERN SELECT MODE

Row A + Row B become 16-slot pattern selector.
LED states encode pattern status across all 16 slots.

Row A — patterns 1–8

1	2	3	4	5	6	7	8

Row B — patterns 9–16

9	10	11	12	13	14	15	16

Empty slot · Active (ON) · Queued (BLINK)

Release SW2 to load queued pattern on next bar boundary — seamless live switch aligned to OXI transport. Pattern LEDs provide persistent status: no screen interaction needed during performance.

OLED 4-LINE DISPLAY ARCHITECTURE · 128×64px


```
BPM:128.0  SW:18%  [PLAY]

TRK:BD  PG:1/2  PROB:75%

LVL  PIT  DEC  PAN  FIL  REV

DLY  PRB  ·  [SHIFT: OFF]
```

8-voice level meters

LINE 1 — GLOBAL TRANSPORT

BPM, swing %, playback state. Always visible regardless of mode. Updated every MIDI clock tick.

LINE 2 — ACTIVE VOICE + PAGE

Selected voice name, step page (1/2), probability value. Updates on voice select or SW2 tap.

LINE 3 — K1–K6 ASSIGNMENT REMINDER

Abbreviated param names for K1–K6. Switches to shift-mode names on SW1 hold.

LINE 4 — K7–K8 + MODE STATE

K7–K8 labels + current shift state indicator. Eliminates need to memorize shift layer.

MIDI CLOCK INTEGRATION — TEMPO-LOCKED LED FEEDBACK

OXI ONE MK1
Master clock
24ppq MIDI

TRS MIDI

DAISY FIELD MIDI IN
Parse clock
& Note On

24ppq + NoteOn

LED HANDLER (firmware)
blink_phase[]
flash_timer[]

```
// Called in MidiHandler — runs at interrupt priority

void OnMidiMessage(MidiEvent msg) {
  if (msg.type == TimingClock) {
    if (++midi_tick % 6 == 0)      // 16th note
      blink_phase = !blink_phase;
  }
  if (msg.type == NoteOn) {
    int v = NoteToVoice(msg.data[0]);
    if (v >= 0)
      flash_timer[v] = 80;         // 80ms flash
  }
}
```

```
// In main loop: key_state[v] = flash_timer[v]>0 ? LED_ON : (muted[v] ? LED_BLINK : LED_ON)
```

BLINK SYNC

tick % 6 == 0 → toggle blink_phase
= 16th-note tempo-locked blink

TRIGGER FLASH

NoteOn → flash_timer[v] = 80ms
Returns to persistent state after

FULL CONTROL MAP — INFORMATION DENSITY SUMMARY

Control	Normal layer (SW1 off)	SW1-shift layer (hold)	SW2 held (pattern)
K1	Level — voice amplitude	BPM — master tempo	—
K2	Pitch / tune	Swing amount	—
K3	Decay time	Master volume	—
K4	Pan L–R	Pattern length	—
K5	Filter cutoff	Fill probability	—
K6	Reverb send	Clock divider	—
K7	Delay send	Gate length	—
K8	Step probability	Accent level	—
Row A (×8)	Voice select BD–FX	Mute / unmute toggle	Pattern select 1–16
Row B (×8)	Steps 1–8 on/off edit	Steps 9–16 on/off	Steps 9–16 on/off
SW1	Hold = shift active	—	—
SW2	Tap = page 1↔2 flip	Steps 9–16 context	Hold = pat. select
SW1 + SW2	Save current pattern	—	—

13 physical controls

3 interaction layers

24+ unique functions

3 LED states / key

Key Takeaways

01 OXI One absorbs all sequencing complexity — freeing every Daisy Field key for performance state display rather than step entry.

02 3-state LED encoding (OFF / BLINK / ON) plus trigger flash delivers 4 distinct voice states per key with zero additional hardware.

03 Shift-hold (SW1) doubles knob parameter space without mode-lock — layer is physical, not logical, so muscle memory is preserved.

04 MIDI clock sync from OXI makes blink tempo-musical — muted voice indicators pulse in time, turning a UI cue into a performance aid.

05 OLED as persistent cheat sheet — 4-line format eliminates the need to memorize shift functions, reducing cognitive overhead mid-performance.